CLINICAL · NOSE & SINUSES

Nose and sinus operations: principles

Nose and sinus operations are best learned by what each one opens, removes and risks. The old operations (antral puncture, inferior meatal antrostomy, Caldwell-Luc, submucous resection) are still examined and occasionally used; endoscopic sinus surgery has replaced most of them by restoring drainage through the natural openings. The step-by-step guides are in Theatre.

AT A GLANCE

- Septoplasty has replaced submucous resection.
- Endoscopic sinus surgery restores drainage through the natural ostia.
- Major complications: orbital injury, CSF leak, carotid injury.
- Know the danger areas: lamina papyracea, cribriform plate, anterior ethmoidal artery.
- Appropriateness: Lund–Mackay ≥1, 8 weeks of maximal medical therapy, SNOT-22 ≥20.

1 CHAPTER ONE Overview

From puncture to endoscope.

1.1 The operations

NOSE AND SINUS OPERATIONS

- DIAGNOSTIC **Nasal endoscopy** antral puncture (proof puncture)
- SEPTUM AND TURBINATES **Septoplasty** submucous resection, turbinate reduction
- MAXILLARY SINUS **Middle meatal antrostomy** inferior meatal antrostomy, Caldwell-Luc
- ETHMOID, FRONTAL, SPHENOID **Endoscopic sinus surgery** external ethmoidectomy, frontal trephine, Lynch-Howarth, osteoplastic flap, Draf procedures
- SKULL BASE **Endoscopic skull base surgery** CSF leak repair, pituitary surgery
- OTHER **Oroantral fistula repair** balloon sinuplasty

1.2 The danger areas

The lamina papyracea (orbit), the cribriform plate and its thin lateral lamella (CSF leak), the anterior ethmoidal artery in the skull base (orbital haematoma), the optic nerve and carotid artery in the sphenoid and Onodi cells, and the nasolacrimal duct in front of the maxillary ostium.

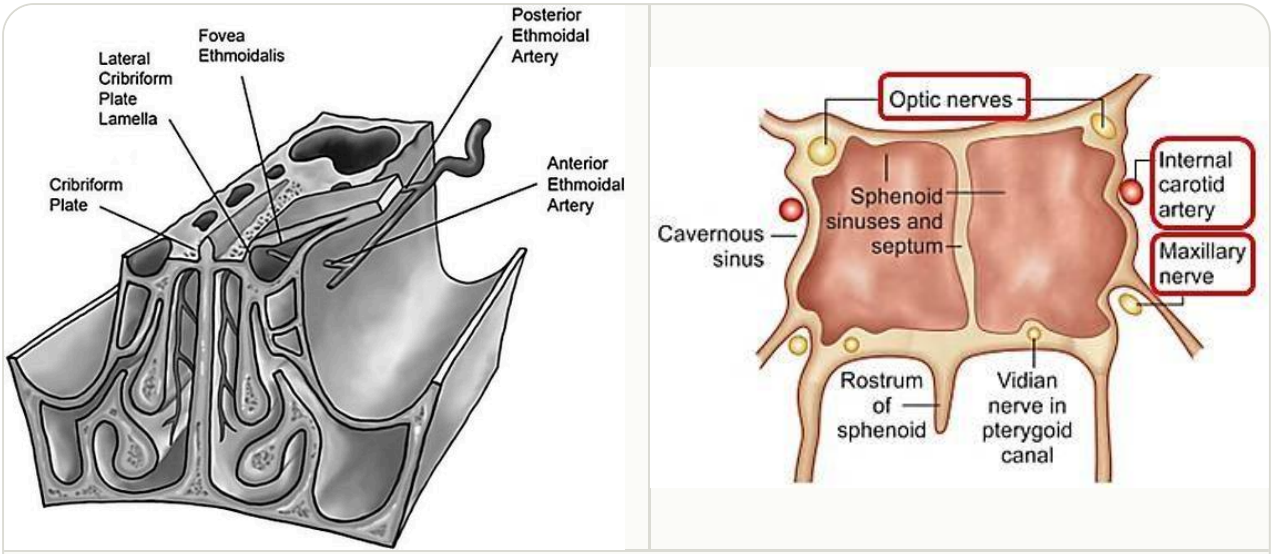


FIGURE 1 The ethmoid roof and anterior ethmoidal artery, and the relations of the sphenoid sinus: optic nerve, internal carotid artery and vidian nerve.

Anatomy atlas · ENT Steps p. 139, ENT Steps p. 154

2 CHAPTER TWO Diagnostic nasal endoscopy

The modern examination.

2.1 Indications

Every chronic nasal complaint: obstruction, discharge, epistaxis, smell loss, headache, suspected tumour or CSF leak, and follow-up after surgery; biopsies and swabs under vision.

2.2 Technique

After topical decongestant and anaesthetic (4% lignocaine with a vasoconstrictor), with the patient sitting or lying, a 4 mm 0° or 30° rigid endoscope makes three

passes:

- **First pass:** along the floor to the nasopharynx: inferior turbinate and meatus, Eustachian tube, fossa of Rosenmüller.
- **Second pass:** between the middle and inferior turbinates to the sphenoethmoidal recess and superior meatus.
- **Third pass:** under the middle turbinate into the middle meatus: uncinate process, bulla, hiatus semilunaris and accessory ostia.

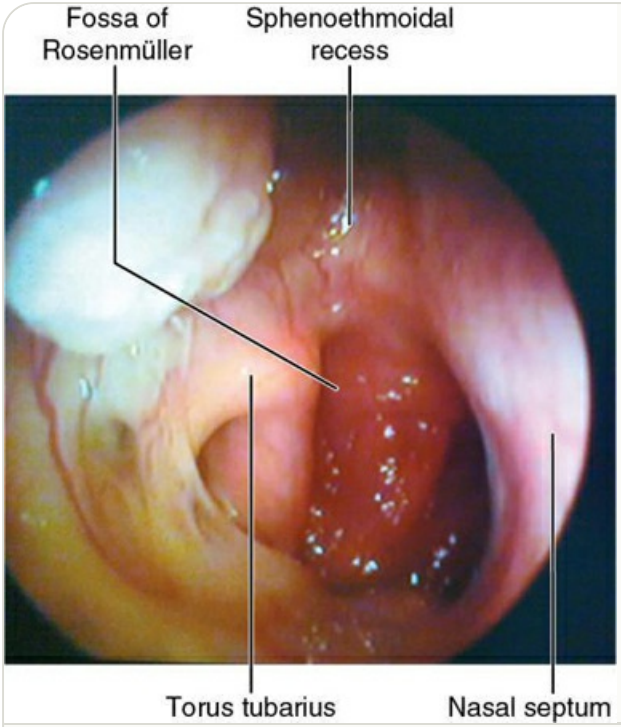


FIGURE 2 The region of the Eustachian tube, sphenoethmoidal recess, choana and fossa of Rosenmüller.

Dhingra, fig. 89.1

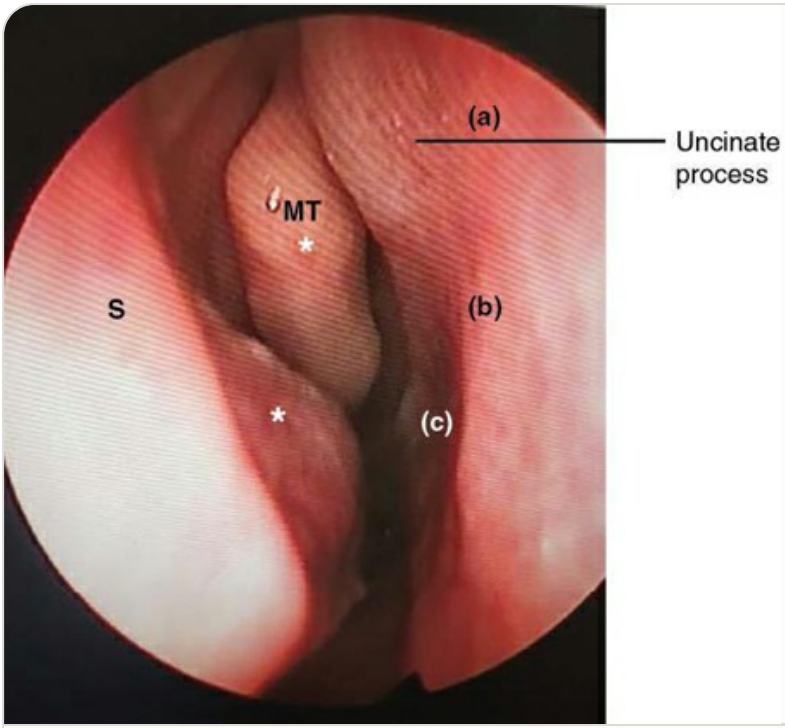


FIGURE 3 Endoscopic view of the left nasal cavity showing the uncinate process, with sites for injection on the lateral wall and middle turbinate.

Dhingra, fig. 90.2

CHAPTER THREE

3 Old maxillary operations

Still examined.

3.1 Antral puncture (proof puncture)

3.1.1 Indications

Chronic or subacute maxillary sinusitis, to confirm the diagnosis and wash out pus; to obtain a specimen for culture or cytology. Contraindicated in acute sinusitis for fear of osteomyelitis.

3.1.2 Technique

Under local anaesthesia, a Lichtwitz trocar and cannula pierce the thin lateral wall of the inferior meatus 1.5 to 2 cm behind the front of the inferior turbinate, directed towards the ear on the same side; the sinus is irrigated with warm saline until the return is clear.

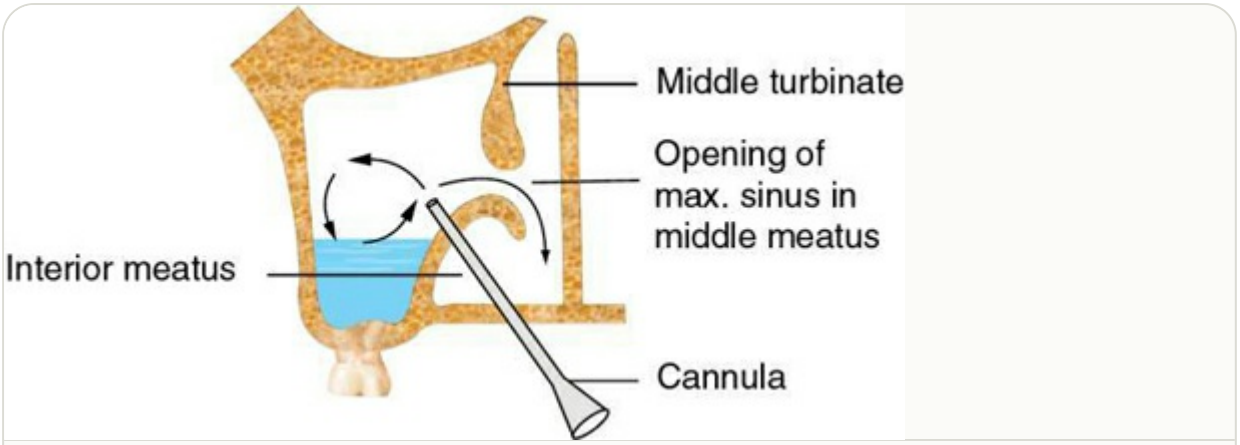


FIGURE 4 Antral puncture through the inferior meatus.
Dhingra, fig. 84.1

3.1.3 What comes back

Thin amber fluid with cholesterol crystals means a cyst; blobs of mucopus mean hyperplastic sinusitis; foul pus mixing with the fluid means suppuration.

3.1.4 Complications

Cheek swelling (the cannula in the soft tissues), orbital injury through the roof, puncture of the posterior wall, bleeding and, rarely, fatal *air embolism* (never blow air into the antrum).

3.2 Inferior meatal antrostomy

A window under the inferior turbinate for drainage. It fails because mucociliary clearance carries mucus to the natural ostium, not to the new opening; risks bleeding and nasolacrimal duct injury. Now obsolete.

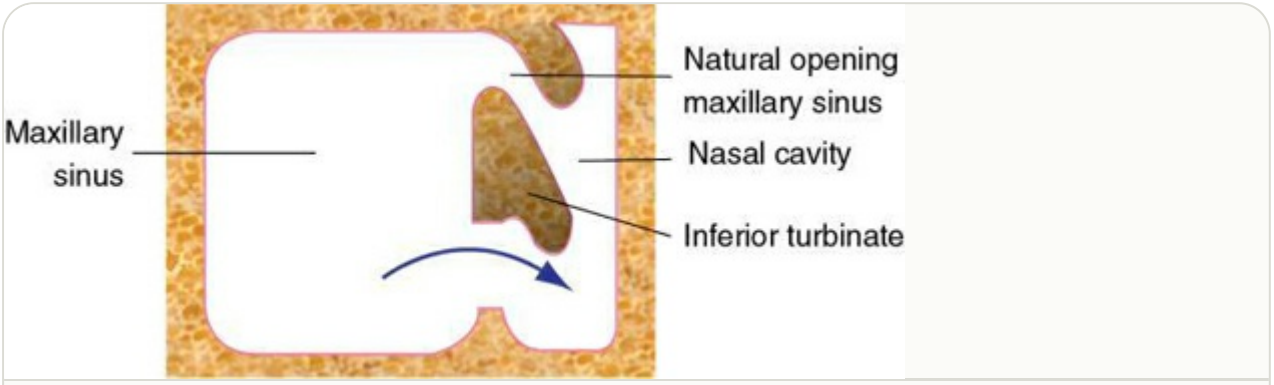


FIGURE 5 Intranasal antrostomy in the inferior meatus.
Dhingra, fig. 85.1

3.3 Caldwell-Luc operation

3.3.1 What it is

Opening the maxillary sinus through its anterior wall in the canine fossa via a sublabial incision, with a nasoantral window.

3.3.2 Indications (historical and current)

Irreversible sinus disease, foreign bodies and tooth roots, dental cysts, oroantral fistula, biopsy of antral tumours, recurrent antrochoanal polyp, maxillary and orbital floor fractures, access to the ethmoids (Horgan) or pterygopalatine fossa (maxillary artery ligation, vidian neurectomy). Avoided in children under about 17, whose developing teeth lie in the anterior wall.

3.3.3 Complications

Bleeding, numbness of the cheek and teeth (infraorbital and superior alveolar nerves), nasolacrimal duct injury, sublabial (oroantral) fistula and, rarely, osteomyelitis.

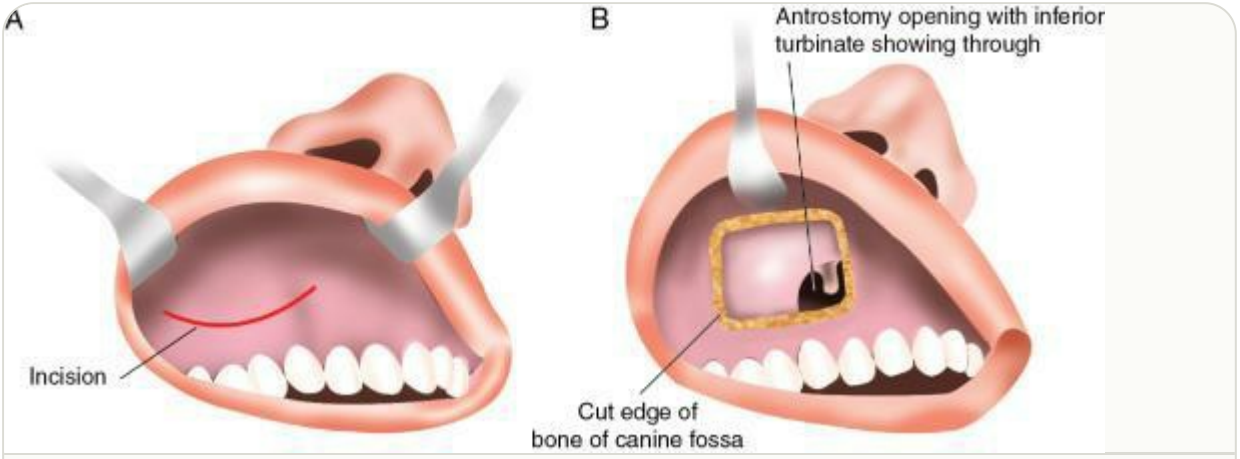


FIGURE 6 The Caldwell-Luc operation: (A) incision, (B) inside view and position of the antrostomy.

Dhingra, fig. 86.1

3.3.4 Present status

Most indications are now handled endoscopically; it remains useful for disease in the anterior and lateral recesses of the sinus and combined approaches.

4 CHAPTER FOUR Septal and turbinate surgery

The airway.

4.1 Submucous resection

Flaps raised on both sides of the septum and all deformed cartilage and bone removed, leaving only dorsal and caudal struts. Not done before about 17.

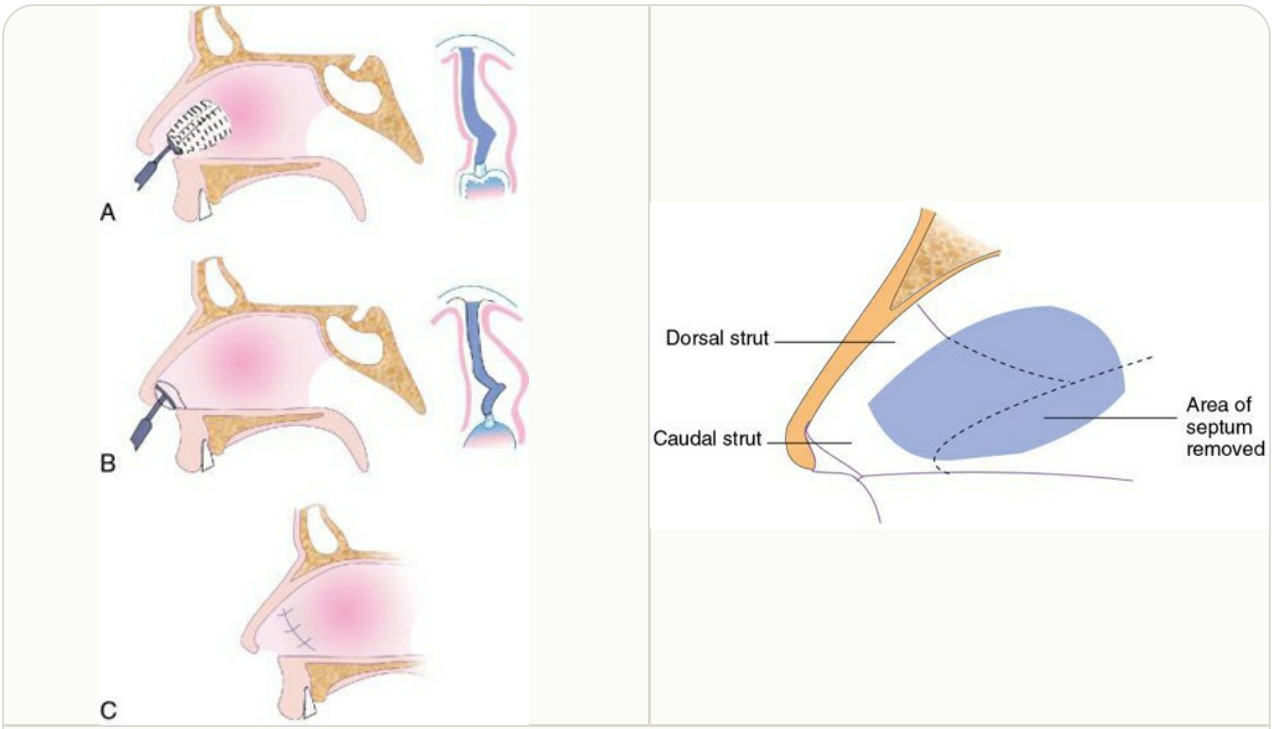


FIGURE 7 First, submucous resection: incision and elevation of the flaps. Second, the dorsal and caudal struts that must be preserved to avoid a supratip depression.

Dhingra, figs 87.1 and 87.2

4.2 Septoplasty

The conservative operation that has replaced submucous resection: a Killian (or hemitransfixion for caudal deviation) incision, a flap on one side, limited removal of bone and cartilage, and correction of the rest by scoring, cross-hatching, shaving or wedge excision and repositioning on the maxillary crest; can be done in children.

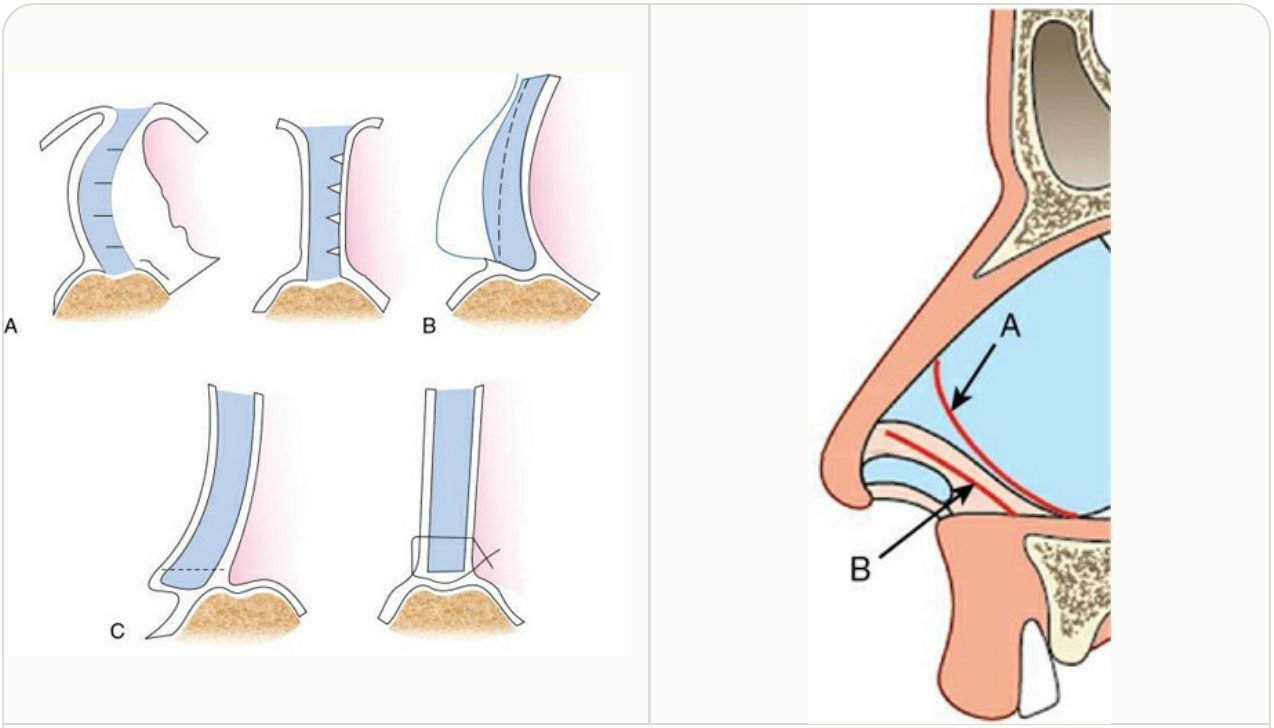


FIGURE 8 First, straightening septal cartilage by scoring the concave side, shaving the convex side, or repositioning a dislocated cartilage. Second, septal incisions: (A) Killian's, (B) hemitransfixion.

Dhingra, figs 88.1 and 88.2

SUBMUCOUS RESECTION OR SEPTOPLASTY?

	SUBMUCOUS RESECTION	SEPTOPLASTY
Dissection	Extensive, flaps on both sides	Limited, one flap
Cartilage	Removed, leaving dorsal and caudal struts	Mostly preserved and repositioned
Age	Not before about 17	Children too
Complications	Supratip depression, columellar retraction, flapping septum, perforation	Fewer
Revision	Difficult	Easier

4.3 Indications for septoplasty

Obstruction from a deviated septum, as part of septorhinoplasty, epistaxis from a spur, sinusitis or contact headache from deviation, access for endoscopic sinus, lacrimal or pituitary surgery, and sleep-disordered breathing.

4.4 Complications of septal surgery

Bleeding, septal haematoma and abscess, perforation, saddle nose and supratip depression, columellar retraction, synechiae, anosmia, and rarely CSF leak.

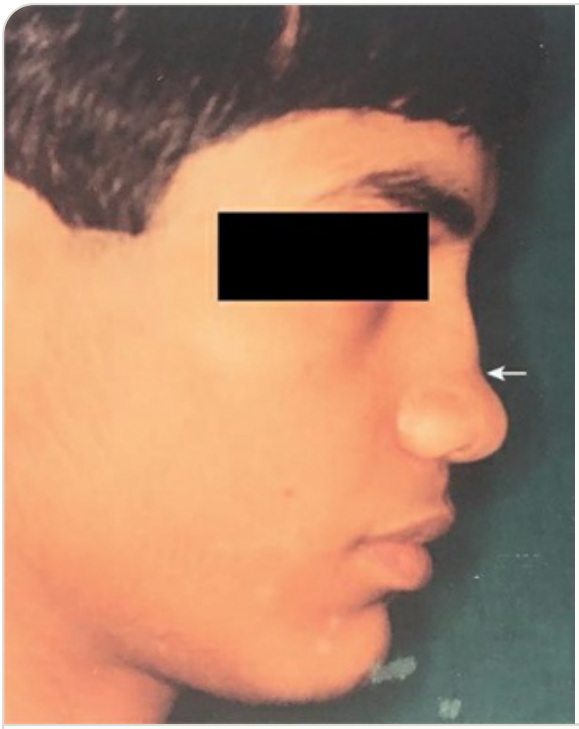


FIGURE 9 A supratip depression after extensive septal surgery.
Dhingra, fig. 87.3

4.5 Turbinate surgery

Submucosal reduction (microdebrider, radiofrequency, coblation), outfracture, partial turbinectomy, laser; preserve mucosa to avoid an empty nose.

5 CHAPTER FIVE Endoscopic sinus surgery

Restoring drainage.

5.1 Principle

Functional endoscopic sinus surgery removes disease in the ostiomeatal complex and enlarges the natural ostia, so that ventilation and mucociliary clearance

recover; the mucosa is preserved.

5.2 Indications

Chronic rhinosinusitis with or without polyps after failed medical treatment, recurrent acute sinusitis, complications of sinusitis (orbital abscess, mucocele), fungal sinusitis, antrochoanal polyp, and access for extended procedures: CSF leak repair, orbital and optic nerve decompression, dacryocystorhinostomy, sphenopalatine artery ligation, choanal atresia, tumours and the skull base.

5.3 Preparation

CT of the sinuses reviewed systematically before every operation (the *CLOSE* checklist: cribriform plate, lamina papyracea, Onodi cell, sphenoid and carotid, ethmoidal artery); topical decongestion and infiltration.

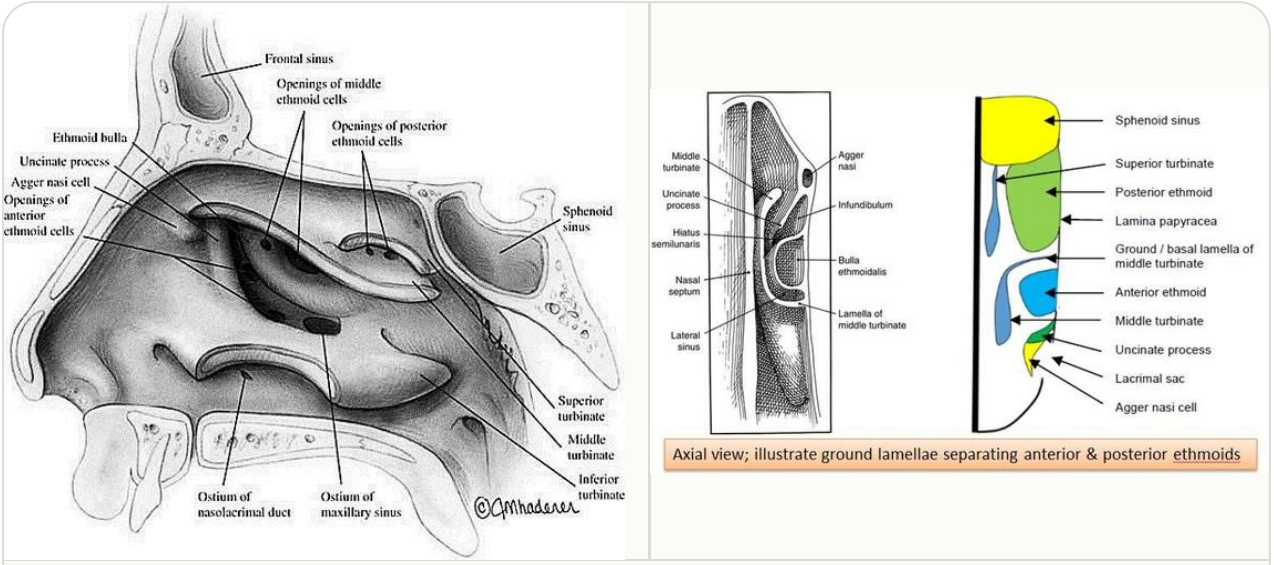


FIGURE 10 The lateral nasal wall with the middle turbinate removed (uncinate, bulla and hiatus), and the ground lamella separating anterior from posterior ethmoid cells.

Anatomy atlas · ENT Steps p. 108, ENT Steps p. 142

5.4 Steps (anterior to posterior, Messerklinger-Stammberger)

1. Medialise the middle turbinate; trim a concha bullosa by removing its lateral lamella.
2. **Uncinectomy.**

3. **Middle meatal antrostomy:** find and enlarge the natural maxillary ostium, which lies behind the lower uncinate, above the inferior turbinate.
4. **Bullectomy** (anterior ethmoidectomy), staying away from the orbit, skull base and anterior ethmoidal artery.
5. **Posterior ethmoidectomy** through the lower, medial part of the basal (ground) lamella; beware an Onodi cell around the optic nerve.
6. **Frontal recess clearance and frontal sinusotomy** if diseased: the opening lies lateral to the middle turbinate attachment, medial to the orbit, in front of the anterior ethmoidal artery and behind the agger nasi; respect the mucosa to prevent stenosis.
7. **Sphenoidotomy** through the natural ostium, about 1 cm above the choana, medial to the superior turbinate.

The posterior-to-anterior (*Wigand*) technique starts at the sphenoid and follows the skull base forwards, for extensive polyposis and revisions.



FIGURE 11 Endoscopic surgery in progress, with instruments passed through the nose and the view on a monitor.
Dhingra, fig. 90.1

5.5 Frontal sinus procedures

Draf I (frontal recess clearance), Draf IIa and IIb (enlarging the frontal ostium), and Draf III (the endoscopic modified Lothrop, joining both frontal sinuses across the midline). External options: trephination, Lynch-Howarth frontoethmoidectomy, the osteoplastic flap with obliteration, and cranialisation after posterior wall fractures.

5.6 Balloon sinuplasty

Dilatation of the natural ostia with a balloon catheter, for selected patients with limited disease.

5.7 After surgery

Saline irrigation, steroid sprays for polyposis or allergy, antibiotics as needed, and endoscopic cleaning of crusts and division of adhesions at weekly visits until the cavity heals.

5.8 Complications

COMPLICATIONS OF ENDOSCOPIC SINUS SURGERY

TYPE	EXAMPLES
Minor	Synechiae, crusting, bleeding, middle turbinate lateralisation, periorbital bruising, surgical emphysema of the lid, dental pain
Orbital	Breach of the lamina papyracea, orbital haematoma (anterior ethmoidal artery), injury to the medial rectus, nasolacrimal duct injury and epiphora, optic nerve injury and blindness
Intracranial	CSF leak, meningitis, pneumocephalus, brain injury
Vascular	Internal carotid artery injury, sphenopalatine and ethmoidal haemorrhage
Other	Anosmia, frontal sinus stenosis and mucocele

ORBITAL HAEMATOMA DURING SURGERY

Proptosis, a tense eye and falling vision: stop, remove packs, and perform a lateral canthotomy and cantholysis; consider medical treatment and orbital decompression.

OPERATIVE GUIDES IN THEATRE 16 operations

Office-Based Diagnosis of Sinonasal Disorders
Office endoscopy

Septoplasty—Classic and Endoscopic
Septoplasty

Surgery of the Turbinate
Turbinates

Balloon Sinus Dilation
Balloon dilation
Antral Lavage; Endoscopic Middle Meatal Antrostomy/Antrochoanal Polyp
Antrostomy
Anterior Antrostomy: The Caldwell-Luc Operation
Caldwell-Luc
Oroantral Fistulas
Oroantral fistula
Endoscopic Ethmoidectomy
Ethmoidectomy
External Ethmoidectomy
External ethmoidectomy
Endoscopic Sphenoidotomy
Sphenoidotomy
Trephination of the Frontal Sinus
Frontal trephination
Endoscopic Approach to the Frontal Sinus
Endoscopic frontal sinus
Draf III, Endoscopic Modified Lothrop
Draf III
Osteoplastic Obliteration of the Frontal Sinus
Osteoplastic flap
Frontal Sinus Cranialization
Cranialisation
Endonasal Approach to the Sella and Parasellar Areas
Endonasal pituitary surgery

CHAPTER SIX

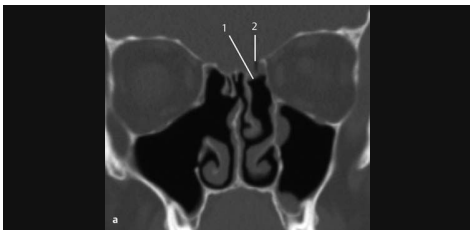
6 Endoscopic skull base surgery

Through the nose to the brain.

Extended endonasal approaches reach the sella (pituitary tumours), the anterior skull base and cribriform plate (olfactory neuroblastoma, meningioma), the clivus and the pterygopalatine and infratemporal fossae. The defect is closed in layers, and the *nasoseptal flap* on the posterior septal artery has greatly reduced postoperative CSF leaks (see the CSF rhinorrhoea topic).

IMAGING IN THE APP 3 cases

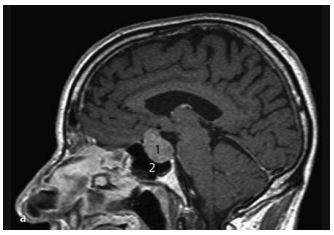
A dural defect after sinus surgery, a frontal recess mucocele, and pituitary pathology.



Dural Defect after Sinonasal Surgery
After surgery · 3 images



Mucocele of the Frontal Recess
Infection & inflammation · 3 images



Pituitary Gland Pathology
Tumours · 3 images

CHAPTER SEVEN

7 Oroantral fistula

A hole between mouth and sinus.

7.1 Causes

Extraction of the second premolar or upper molars, whose roots lie close to the sinus floor (especially with an apical abscess); a Caldwell-Luc incision that fails to heal; carcinoma; fractures; osteitis, syphilis and granulomatous disease.

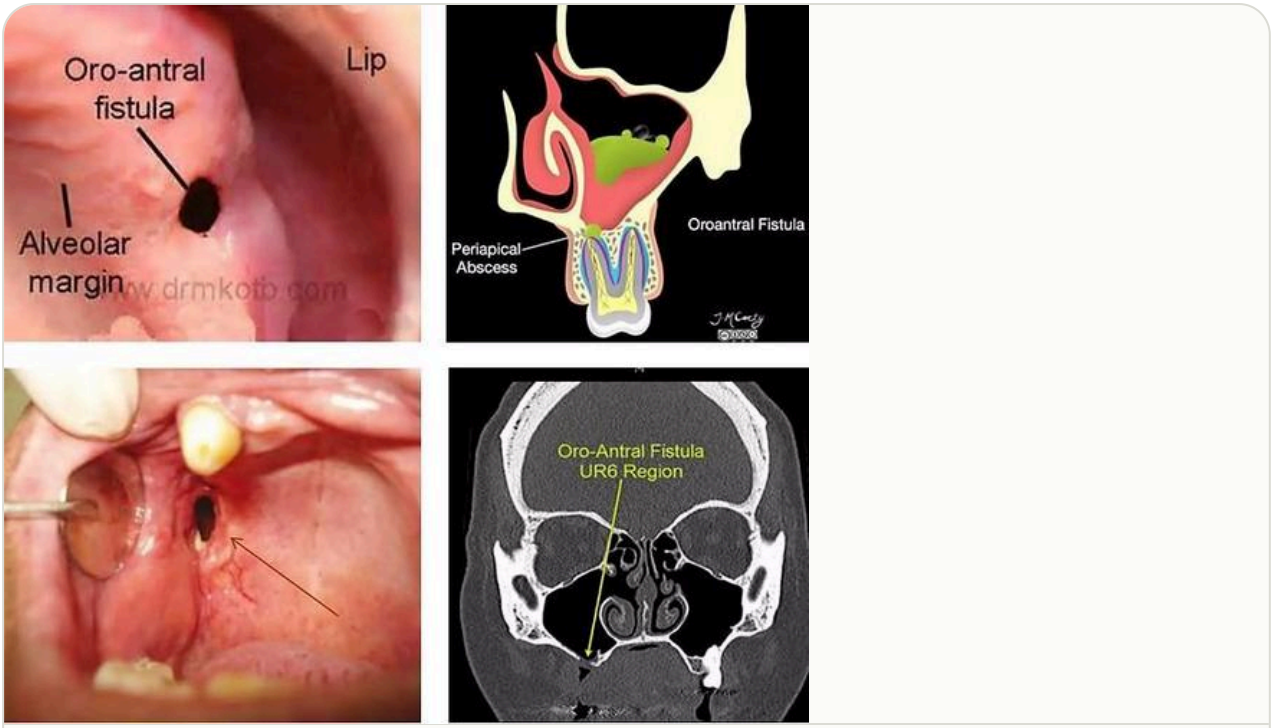


FIGURE 12 An oro-antral fistula: the thin floor of the maxillary sinus lies over the roots of the upper teeth.

Anatomy atlas · ENT Steps p. 132

7.2 Clinical picture

Fluids and food regurgitate into the nose; a foul discharge from the infected sinus into the nose or mouth; the patient cannot drink through a straw or blow a wind instrument. A probe passes from the socket into the antrum.

7.3 Treatment

- **Fresh, small fistula** without infection or a retained root: suture the gum margins and give antibiotics.
- **Chronic or large fistula:** treat the sinusitis (endoscopic antrostomy, removal of any root), excise the epithelialised track and close it with a buccal advancement or palatal rotation flap; or an obturator after cancer surgery.

SELECTION, OUTCOMES AND REVISIONS (OXFORD TEXTBOOK)

Who should have surgery. An international appropriateness consensus (RAND method) defined a minimum below which endoscopic sinus surgery is unlikely to help: in adults with uncomplicated chronic rhinosinusitis, a CT Lund–Mackay score of at least 1, a trial of at least eight weeks of topical nasal steroid plus a short course of

systemic steroid (with polyps) or antibiotics (without polyps), and a SNOT-22 score of 20 or more after that treatment. Up to one in three patients do not achieve a clinically meaningful improvement after surgery, and the **preoperative SNOT-22** is the best predictor of benefit, so selection matters. Equally, repeated rounds of medical treatment that delay surgery in patients who meet the criteria may worsen the final result.

Technique. Revision cases most often show a residual uncinate process, a missed natural ostium with a separate antrostomy (causing mucus recirculation) and residual anterior ethmoid cells, so the basic steps done well prevent many revisions. Polyp disease does better with more complete surgery (wide antrostomies, clearing all ethmoid partitions, extended frontal and sphenoid surgery) to give topical steroids access. Balloon dilatation cannot treat the ethmoids and suits only limited disease without polyps. Total intravenous anaesthesia gives a drier field; routine preoperative steroids and antibiotics are not needed. Image guidance is useful in revisions and abnormal anatomy, but has not been shown to reduce complications, and over-reliance can mislead. Steroid-eluting stents may help selected patients.

Revision rates are about 11% at three years, 19% at five and 17% at ten. A patient who develops proptosis during or after surgery needs urgent assessment of vision, pupils and intraocular pressure, and immediate decompression of an orbital haematoma (lateral canthotomy and cantholysis) if pressure is raised.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Antral puncture:** site, direction, findings and complications, including air embolism.
- **Caldwell-Luc operation:** incision, indications, complications, why not in children.
- **Submucous resection versus septoplasty.**
- **Steps of endoscopic sinus surgery** and the danger areas.
- **Complications** of endoscopic sinus surgery and their management.

- **Oroantral fistula.**

8.2 Common pitfalls

COMMON PITFALLS

- Operating without reviewing the CT for the dangerous variants.
- Making a second maxillary opening that does not join the natural ostium (recirculation).
- Stripping mucosa in the frontal recess.

Written for this app from Oxford Textbook of Otolaryngology ch. 37, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 20, Dhingra ch. 84 to 90 and K. J. Lee ch. 33, 34. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

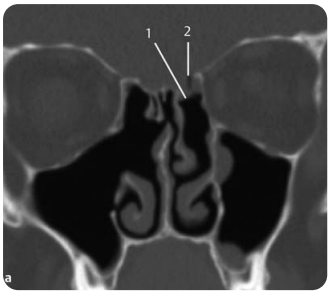
On imaging

The imaging of this topic: key findings, differentials and representative scans.

PARANASAL SINUSES · AFTER SURGERY

Dural Defect after Sinonasal Surgery

- The commonest site of iatrogenic skull base injury is the lateral lamella of the cribriform plate (thinnest bone, where the anterior ethmoidal artery enters), then the posterior ethmoid roof.
- Locate the defect with thin-section coronal CT; MR cisternography (heavily T2-weighted) shows CSF tracking through it; beta-2 transferrin confirms leaked fluid.
- Small defects are repaired endoscopically with grafts; larger ones with a vascularised (nasoseptal) flap.



CT, coronal. This slice at the level of the olfactory fossa shows a bone defect involving the left cribriform plate (1), indicating the location of th…

PARANASAL SINUSES · INFECTION & INFLAMMATION

Mucocele of the Frontal Recess

- Beware of large osseous lesions of the skull base, removal of which may result in protrusion of intracranial contents or marsupialization of the mucocele.
- In meningo(encephalo)cele or intracranial tumors, persistent leakage of cerebrospinal fluid may occur at biopsy.



CT with intravenous contrast enhancement, axial. The lesion in the anterior skull base (1) in the area of the frontal recess shows signs of…